

# Experiment - 11

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Semester: 5  
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Date of Performance: 21/9/26  
Subject Code: 24CSP-310

## Q1.

A company stores the monthly salary history of its employees in an EmployeeSalaryHistory table. Write an SQL query to find the employees whose salary has increased in every recorded month compared to the immediately previous month. For each salary record, compare the current month's salary with the employee's previous month's salary using a Window Function. Display only those employees for whom all later salary records are greater than their respective previous salary.

The output should display:

- Employee ID
- Employee Name

Output:

| employee_id | employee_name |
|-------------|---------------|
| 101         | Alok          |
| 103         | Raj           |

## SOLUTION:

```
SELECT employee_id, employee_name
FROM (
    SELECT *,
        LAG(salary) OVER(
            PARTITION BY employee_id
            ORDER BY salary_month
        ) prev_salary
    FROM EmployeeSalaryHistory)
GROUP BY employee_id, employee_name
HAVING COUNT(*) FILTER (
```

WHERE salary <= prev\_salary  
 ) = 0;

## Q2.

A company maintains employee details in an Employee table and department information in a Department table. Write an SQL query to find the employees whose salaries belong to the top three distinct salary levels within their respective departments. If multiple employees have the same salary, they should receive the same salary rank and all of them should be included in the result.

The output should display:

- Department Name
- Employee Name
- Salary

Output:

| Department | Employee | Salary |
|------------|----------|--------|
| IT         | Max      | 90000  |
| IT         | Jim      | 85000  |
| IT         | John     | 85000  |
| IT         | Alex     | 75000  |
| Sales      | Henry    | 80000  |
| Sales      | Susan    | 75000  |
| Sales      | David    | 70000  |
| HR         | Tina     | 60000  |
| HR         | Rita     | 55000  |
| HR         | Manoj    | 50000  |

## SOLUTION:

```
select d.name, e.name, e.salary
from (
select *, dense_rank() over (partition by departmentid order by salary desc) as rank
from employee
) e
join department d
on e.departmentid=d.id
where rank<=3;
```